# Revolut Analytical Challenge

You are a member of the team responsible for profitability at Revolut. The team tracks **total gross profit* in the last 90 days (GP L90D)** as its primary metric.

Given Revolut's rapid changes to product offering, the team is exploring whether recent initiatives are negatively impacting GP L90D.

**Gross Profit = Revenue - Cost of Sales*

The team has a dataset of user-level revenues & costs in the last 90 days, with the details explained below.

**Objective:** The team is looking to increase GP L90D by **£4 million** in the next 6-12 months. You have been tasked with providing recommendations to achieve this target based on your findings.

You are free to use Excel or any scripting language to parse and analyse the data.

**Note:** Please show all your work (including your code if applicable), specify any assumptions that you make and ensure that your summary output is in a PDF format (max 10 pages) and presentable to the executive leadership team. Please make sure you share document access in advance.

For reference, the team has also compiled a list of recent initiatives/changes within the company (though not exhaustive). Here is what they know so far:

**Revolut Retail Launches:**

1. Credit team has reduced interest rates on loans. Changes are affecting users in Romania, Germany and Lithuania on all plans.
2. Savings team increased interest rates on its instant access savings accounts for personal customers. Changes have been added across all countries for Metal & Ultra plan holders.
3. Lifestyle team launched E-sims with a third-party vendor. Users can purchase eSim data plans directly in the app for 150+ countries (available to everyone).
4. New Pricing system has been introduced for users of Stays, reducing the amount of cash back that they receive by 30% on each transaction.

5. Bank Payments in HUF and RON had updated pricing schedules, with the first 2 transfers in the month free for Premium plan users (previously all paid).
6. Growth team ran a campaign, which allowed users to pick up cards from vending machines and provide a monetary reward to users after eligibility check. To become eligible for the reward, the user needs to sign up and spend at least £20 (i.e. become a New Active Person)
7. Subscription team rolled out new benefits in Plus & Ultra plans to increase conversion to paid plans. Users are now able to access partner subscriptions for free with these plans, including a video-streaming service, and 2 app-based services for health & wellness.
8. Card payment vendors introduced changes in cost structure for personal customer payments. These are affecting all cost-based services provided by these partners (ATM withdrawals, top-ups of the account)
9. Revolut signed a new vendor for creating and shipping new cards. The vendor now handles 100% of the volume across all countries and all card types.

**Revolut Business Launches:**

10. Business team introduced changes to the onboarding journey simplifying the number of steps needed to create an account for a business. There are no more questions about the number of employees that a Business has and no additional questions relating to countries where a business has suppliers.
11. Acquiring team launched a new terminal for offline payments. This terminal can handle payments processing, managing restaurant orders, and can act as a point-of-sale terminal for small shop owners.
12. Business team launched Money Market Funds for businesses with the highest interest rate on the market. The service is offered in partnership with a well-established asset manager and available to all plans except Basic.
13. Business team launched a new paid plan in March called Scale, aimed at larger businesses with more complex needs, which provides discounted foreign exchange operations.

**Other Initiatives:**

14. Card Payments team improved the refund settlement speed for users in the UK and Ireland. Users can now get a refund on a payment within 3 hours of a merchant confirming the release of money from their account.
15. Legal team has streamlined the process for launching new initiatives at Revolut. There are now 5 less steps required from the Product Manager when submitting the initiative for governance, resulting in a 50% decrease in time needed to get all the necessary approvals.
16. New legislation introduced by the regulator impacting provisions for loans from January 2024. Loans in Germany according to the internal model now have to include up 10% of the face value as provisions based on the quality of the borrower (5% previously)
17. Revolut built an in-house system for bulk sending SMS notifications. This means Revolut can offboard the vendor that automates the process, but still needs to pay to the mobile operators for every SMS sent.

# Dataset Glossary

| Column | Explanation |
|---|---|
| user_id | Unique user identifier. |
| pnl_date | Date for which the relevant revenues and costs are recorded. |
| user_city & user_country | Country and city of residence. |
| account_level_1 - account_level_4 | Profit and loss (P&L) account levels (categories) used for structuring management P&L statements. Each level 1 consists of several level 2s. Each level 2 consists of several L3 and so on. Full explanation of what each of the account levels means can be found below. |
| amount_gbp | Gross profit generated by a user. Measured in £1,000s |
| user_type | Type of customer available:<br>● Personal - refers to personal Revolut App customers<br>● Youth - refers to users of the <18 App<br>● Business - refers to businesses of the Business app<br>● Business_User - refers to a director of the company who owns the account<br>● Business_employee - employee of the company who has access limited access to a business account<br>● Company - refers to Revolut entity |
| age_group | Pre-defined age groups of users. |
| user_state | Can be of two values:<br>● Active - the user has completed all the checks and is ready to use the account<br>● Inactive - Inactive users have restricted access to the app. To become active they need to complete some outstanding checks.<br><br>***Important:*** *Inactive state applies to users who:*<br>● *Haven't completed full onboarding*<br>● *Completed onboarding, but their documents or other information has expired* |
| Onboarded date | Date when the user completed onboarding. |

| | |
|---|---|
| User plan | Revolut has a set of plans for both business and retail customers. Field shows the current plan that the user is on.<br>**Personal:**<br>● Standard - the only free plan<br>● Plus<br>● Premium<br>● Metal<br>● Ultra<br>**Company_X:** refers to plans available to registered entities on Revolut Business app<br>● Basic - only free plan<br>● Grow<br>● Scale<br>● Custom<br>**Freelancer_X:** refers to plans available to sole traders on Revolut Business app<br>● Basic - only free plan<br>● Grow<br>● Scale |
| is_nap | NAP (New Active Person) refers to a user who has spent at least £20 pounds on the account and is eligible for the vending machine reward |

## Account Level Information

Account Level 1:

- Treasury - refers to GP earned from company treasury
- Revolut Business - refers to GP earned from customers of the business app
- Revolut Retail - refers to GP earned from customers of the personal & <18 apps

Account Level 2:

- Interest Income - Interest on balances held within Revolut Accounts. Further breakdowns provide information on which assets are held
- Market Making PnL - GP generated from market making activities (such as trading currencies, commodities, etc.). Further accounts provide information on where the assets are held
- Card Payments - refers to GP generated from operations with payment cards. Further accounts provide information on the types of charges
    - ATM - refers to GP from all cash withdrawal operations. There are some partner charges for processing these transactions & a fair usage fee which is charged to customers who go over the monthly limit of withdrawal amount (limit varies by plan)
    - POS charges - specific fees that are paid for processing business card payments at a till
    - Interchange Fees - refers to fees paid to Revolut by card vendors for transactions completed on Revolut cards
    - Top-up - Revolut customers can top-up their account via a card from a different card provider. For these Revolut is charged processing costs by card vendors & earns some surcharge fee for payments below a specific amount.

- Bank Payments - refers to GP generated from bank transfers (directly from account to another account). Further details provide information on costs to process these payments with partners as well as the revenue earned from charging a fee.
- FX - refers to all GP earned from transactions where currency exchange takes place. Further Account levels specify the type of GP earned.
    - Mark-up refers to GP from weekend charge for currency exchange
    - Fees refers to GP earned from currency exchange above the free limit
    - Spread refers to GP from differences in exchange rates offered to customers at settled at internally for the transaction
- Subscriptions - GP earned from selling subscriptions (plans) to customers. These typically include a monthly fee in exchange for some benefits (e.g insurance on personal plans) & some discounts on other fees within the app (e.g transfers). Further breakdowns specify the GP by plan and additional costs of extra benefits & their partner providers
- Lifestyle - refers to GP earned from other non-banking services available on the personal app.
    - Revpoints refers to GP from the loyalty program where users can purchase points that they can spend on plane tickets & holiday bookings
    - Lounge Key refers to GP from selling an access pass to different airport lounges
    - eSIMs refers to GP from selling eSIMs within the app
    - Stays refers to GP earned from selling hotel bookings on the app
    - Shopping Rewards refers to GP from an affiliate program within Revolut. Users can get discounts in shops, for which Revolut earns a commission. Ads refers to big standalone placements with active placement across the app, and commissions for the passive engagements that stay in a separate hub on the app.
- Merchant Acquiring - refers to GP earned from allowing businesses to accept payments
- Treasury Services - refers to GP earned from Savings & Investment services to business, further accounts specify the product sold.
- Credit - refers to GP earned from credit products. Further breakdown specify the product type and components of the GP (Provisions are reserves for individuals that are at risk of default, interchange fee is same as card payments interchange fee)
- Savings - refers to GP earned from savings products. Further accounts specify the products sold

*Tip: whenever you encounter a word “charge” in the account level - the account level refers to cost of sales, whenever there is a word “fee” in the account level - it refers to revenue.*

# FAQs

1. **Can I get additional data sets from the previous 90 Days so I can compare the GP?**

   We don’t provide any additional data sets for this challenge. For the purpose of this challenge, the provided information is enough to identify the root causes and provide recommendations. If you need to make any assumptions please specify them in your solution.

2. **How much time would it take on average to complete this challenge?**

   It will take about 5 - 6 hours to complete it. Our deadline is 7 days
   since we understand that you probably have other projects you’re working on.

3. **What program should I use to do the task?**
   It's completely up to you.

4. **Can I get a smaller data set? I am unable to process this on my computer.**
   Unfortunately we don't have smaller sets to provide to you, solving this is part of the challenge. Also please make sure that you use one of the latest Excel versions since they support larger sets or you can use Google Sheets. Another solution would be using any scripting language since it uses less memory than Excel.

5. **I see some discrepancies in datasets that are not mentioned in the task, can I get some clarification on how to interpret them?**

   We don't provide any clarifications, please interpret any discrepancies you find in whichever way works best for your solution **but note the data is accurate**.